

Sampling & Statistical Significance

GOVT 10: Quantitative Political Analysis

This chapter moves from single estimates to the uncertainty around them. We distinguish populations from samples, develop the standard error and margin of error, meet Student's t-distribution for small samples, and then build the logic of hypothesis testing and p-values: how we decide whether a pattern is more than noise.

1 Sampling and Uncertainty

Earlier in the course — the experimental design class in Week 3 — we learned how to design experiments. Suppose we have done everything right. We picked a clean design, randomly assigned participants, ran a thoughtful treatment, and measured an outcome. We find that the treatment group has a mean outcome of 52 and the control group has a mean of 48. The difference is 4.

Is that a real effect? Or did we just get lucky with who happened to land in which group?

This question is the hinge between design and inference. A well-designed experiment is necessary, but it is not sufficient. We still need to know whether the patterns we observe are signal or noise. This chapter sets up the conceptual machinery that lets us answer that question. The next class will turn that machinery into formal hypothesis tests.

The same logic applies outside of experiments. Every poll, every survey, every observational study based on a sample faces the same question. We observe something; we want to know whether what we observed reflects something real about the world or just an artifact of which subset of the world we happened to look at. The vocabulary we build here (populations, samples, sampling distributions, standard error, margin of error) is the foundation on which every confidence interval and every p-value rests.

1.1 Why Do Polls Disagree?

Three polls are released the same week. All ask the same question about the same population:

- Poll A: 43% approval
- Poll B: 48% approval
- Poll C: 46% approval

Which one is right? How do we know how much to trust any of them?

The answer, which we will build up in the rest of the chapter, is that no single poll is “right.” Each poll is an estimate based on a sample, and different samples produce different estimates. Our job is to understand how much variation to expect across samples, and to use that variation to figure out how confident we should be in any given estimate.

To answer which poll is “right” — or whether any of them is — we need the mechanics of sampling variation. The next several sections build that machinery.

1.2 Populations vs. Samples

A **population** is the complete group we want to learn about. Examples include all registered voters in the United States, all members of Congress, all citizens of a country, or all candidates who ran in the last election. Populations have true, fixed values for any measure: a true average approval rating, a true proportion supporting a policy, a true level of polarization. These true values are called population parameters, and we typically denote them with Greek letters: μ for the mean and σ for the standard deviation.

A **sample** is the subset of the population we actually observe. A Gallup poll of 1,000 voters is a sample. A study of 500 immigrants is a sample. Samples produce estimates of population parameters, denoted with Latin letters: $\bar{x}$ for the sample mean and s for the sample standard deviation.

Tasting soup. You are cooking a giant pot of soup and want to know if it needs more salt. You stir the pot and taste a spoonful.

- Population = the whole pot
- Sample = the spoonful you taste
- μ = how salty the pot actually is
- $\bar{x}$ = how salty your spoonful tastes

A different spoonful would taste slightly different, but if the pot is well stirred, any spoonful tells you something useful about the whole pot. That is the logic of sampling.

We use samples because populations are usually too large to measure directly. There are more than 240 million eligible voters in the United States. You cannot interview them all. So you interview 1,000 and use them to learn about the rest.

1.3 Every Sample Is Different

The fundamental challenge of inference is that different samples from the same population give different results. Imagine four polling firms each conducting an independent survey of 1,000 voters from the same population on the same day.

	Sample 1	Sample 2	Sample 3	Sample 4
Size	1,000 voters	1,000 voters	1,000 voters	1,000 voters
Estimate	48%	55%	30%	49%

Which estimate is “right”? None of them, technically. Each is a different glimpse of the same underlying population. The variation comes from three sources.

Three Sources of Sampling Variation

Random selection. Each sample includes different people by chance. No two random draws are identical.

Population heterogeneity. People differ. Some are strongly supportive, some strongly opposed, most somewhere in between. Different samples capture different mixes.

Sample size. Smaller samples are more sensitive to which particular individuals get drawn. Larger samples smooth out individual quirks.

The crucial insight is that this variation is not chaos. It is random, but it is also predictable. The whole project of statistical inference rests on understanding how sample estimates vary, and on using that understanding to quantify our uncertainty.

1.4 Simulating Sampling Variation

Let us see this in code. We create a hypothetical population of 100,000 people whose true mean approval is 52% with a standard deviation of 15.

```
set.seed(42)
population <- tibble(
  id = 1:100000,
  approval = rnorm(100000, mean = 52, sd = 15)
)

# Take five samples of size 100
five_samples <- map_dfr(1:5, function(i) {
  s <- sample_n(population, 100)
  tibble(sample_id = i, sample_mean = mean(s$approval))
})

five_samples

# A tibble: 5 x 2
  sample_id sample_mean
  <int>      <dbl>
1         1         52.2
2         2         54.4
3         3         52.4
4         4         51.5
5         5         49.7
```

Each row is a different random sample of 100 people drawn from the same population. The true population mean is 52, but no individual sample lands exactly there. Each one is off by some amount, in some direction. This is sampling variation in action.

1.5 The Sample Mean Is Unbiased

Although any individual sample misses the true value, the sample mean is what statisticians call an **unbiased estimator**. On average, across many samples, the sample mean equals the population mean.

$$E[\bar{x}] = \mu$$

The expression $E[\bar{x}]$ is the expected value (the long-run average) of the sample mean. The equation says: if you took thousands of samples and averaged their means, you would land on the true population mean.

```
# Take 1000 samples of size 100, store their means
sample_means <- map_dbl(1:1000, ~ mean(sample_n(population, 100)$approval))

mean(sample_means) # Average of the sample means
```

```
[1] 51.96734
```

The average of 1,000 sample means lands almost exactly on 52, the true population mean. Individual samples vary, but the variation is unbiased: errors in one direction are balanced by errors in the other.

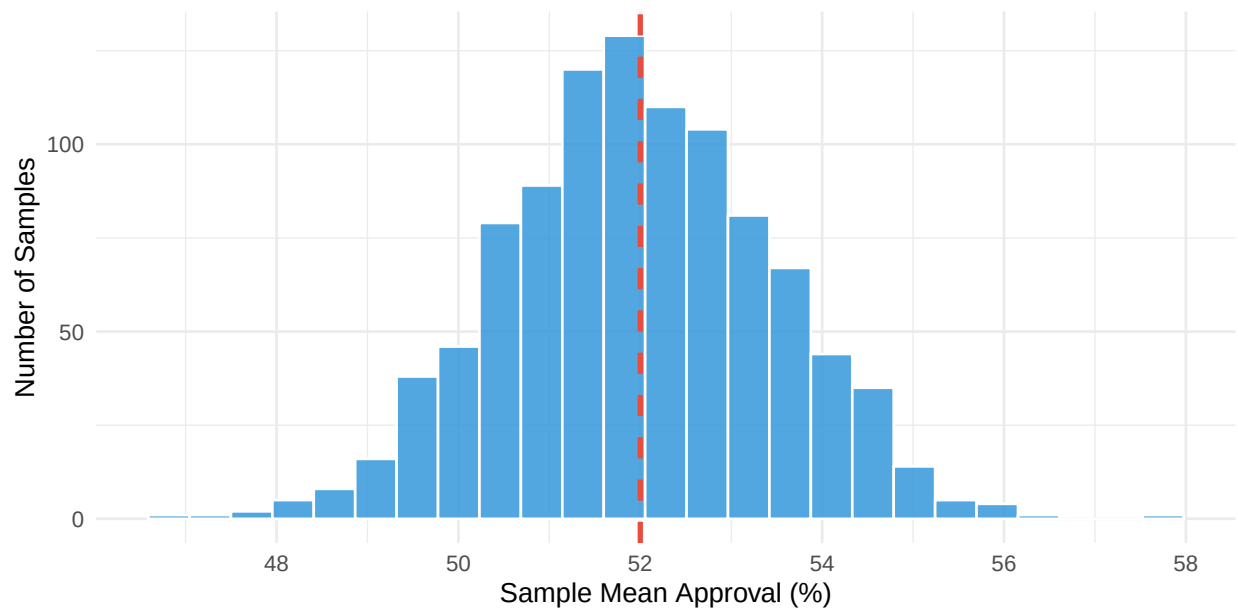
1.6 The Sampling Distribution

Definition

The **sampling distribution** is the set of all possible sample means we would get from repeated sampling of the same size from the same population.

If we plot all those sample means, we see a remarkable pattern. The sample means form a bell-shaped distribution centered on the true population mean.

```
ggplot(sample_means_df, aes(x = sample_mean)) +
  geom_histogram(bins = 25, fill = "#3498db",
                 color = "white", alpha = 0.85) +
  geom_vline(xintercept = 52, color = "#e74c3c",
             linetype = "dashed", linewidth = 1) +
  labs(
    x = "Sample Mean Approval (%)",
    y = "Number of Samples"
  ) +
  theme_minimal()
```



This is the sampling distribution of the mean. Each individual sample produces one estimate, and the collection of those estimates is itself a distribution. Most cluster near the truth (the dashed line at 52), with fewer samples landing far from it.

This pattern is not a coincidence. It is guaranteed by the **Central Limit Theorem**, one of the most important results in statistics. The theorem says that the sampling distribution of the mean approaches a normal distribution as the sample size grows, regardless of the shape of the underlying population. Even if approval ratings in the population are skewed or oddly shaped, sample means will be bell-shaped around the true mean.

This is what makes inference possible. The unpredictability of any individual sample is offset by the predictability of the sampling distribution. We can quantify how spread out sample means are, and use that to gauge how far our particular sample might be from the truth.

1.7 Sample Size and Confidence

Larger samples produce more consistent estimates. This is the law of large numbers in action: with more observations, individual quirks wash out and the sample mean settles closer to the population mean.

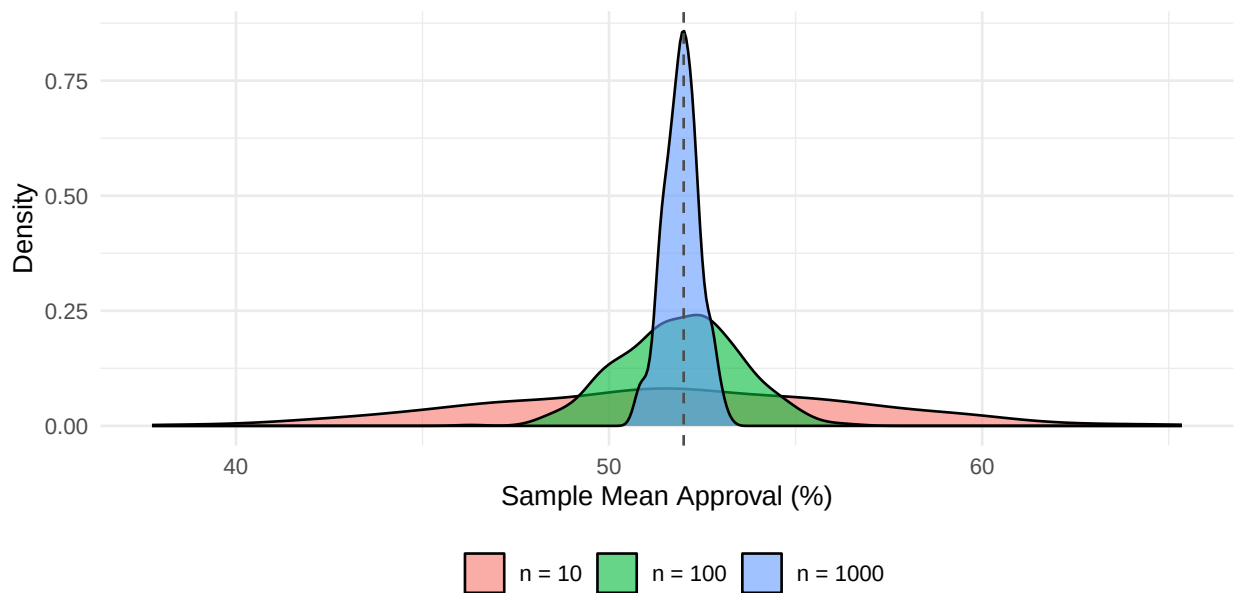
```
# Compare sampling distributions for different sample sizes
size_comparison <- map_dfr(c(10, 100, 1000), function(n) {
  means <- map_dbl(1:500, ~ mean(sample_n(population, n)$approval))
  tibble(sample_size = paste0("n = ", n), sample_mean = means)
})

ggplot(size_comparison, aes(x = sample_mean, fill = sample_size)) +
  geom_density(alpha = 0.6) +
  geom_vline(xintercept = 52, linetype = "dashed", color = "gray30") +
  labs(
    x = "Sample Mean Approval (%)",
```

```

y = "Density",
fill = ""
) +
theme_minimal() +
theme(legend.position = "bottom")

```



All three sample sizes are unbiased: each distribution is centered on 52. But the spread shrinks dramatically as the sample size grows. With $n = 10$, sample means can land anywhere from 40 to 65. With $n = 1,000$, almost all samples land within a percentage point or two of the truth.

This is why pollsters report sample sizes. A larger sample is not just “more data.” It is a tighter sampling distribution, which means more confidence that the estimate is close to the true value.

Notice what is changing and what is not. The center of the distribution does not move with sample size: every sample size produces an unbiased estimate of 52% on average. What changes is the precision around that center. A sample of 10 might land anywhere between 40 and 65. A sample of 1,000 almost always lands within a percentage point or two of 52. The estimate is not getting more accurate in some philosophical sense; it is getting more reliable because each sample is now a better representation of the underlying population.

1.8 Standard Error

We need a single number to summarize how spread out the sampling distribution is. That number is the **standard error**.

$$SE = \frac{\sigma}{\sqrt{n}}$$

Here, σ is the population standard deviation (often estimated from the sample standard deviation s), and n is the sample size. The standard error is the typical distance between a sample mean and the population mean.

Worked example. Suppose the population standard deviation of approval ratings is $\sigma = 15$, and our poll has $n = 100$.

$$SE = \frac{15}{\sqrt{100}} = \frac{15}{10} = 1.5$$

A typical sample of 100 will produce a mean that is about 1.5 percentage points away from the true population mean.

The formula has an important implication: precision improves with the square root of sample size, not with sample size itself. To cut the standard error in half, you have to quadruple the sample.

The Square Root Law

n	$\sqrt{n}$	SE = $15/\sqrt{n}$
100	10	1.50
400	20	0.75
900	30	0.50
1,600	40	0.375

Doubling precision means quadrupling sample size. This is why most national polls stop around $n = 1,000$. Beyond that, the marginal precision gain is rarely worth the marginal cost.

1.9 Standard Deviation vs. Standard Error

These two terms get confused constantly. They are different things.

- **Standard deviation (σ or s)** measures variability among individuals in the population or sample. It answers: how different are people from each other?
- **Standard error (SE)** measures variability among sample estimates. It answers: how different would my estimate be if I drew a different sample?

A population can have a huge standard deviation (people differ a lot) and still produce sample means with a small standard error (the average of a big sample is stable). The two are linked by the formula $SE = s/\sqrt{n}$, but they describe different sources of variability.

When to use which:

- Use **SD** when describing how spread out the data themselves are (e.g., "incomes ranged widely, $s = \$22,000$ ").

- Use **SE** when describing how precise an estimate is (e.g., “the mean income was estimated to within $\pm \$700$ ”).

1.10 Margin of Error

The standard error is a measure of typical variation. The **margin of error** turns that into a range we can report alongside an estimate.

$$\text{MoE} = 1.96 \times SE = 1.96 \times \frac{\sigma}{\sqrt{n}}$$

The 1.96 comes from the normal distribution. About 95% of the area under a normal curve lies within 1.96 standard deviations of the mean. Multiplying the standard error by 1.96 gives us a range that, by construction, will contain the true population value about 95% of the time across repeated samples.

Worked example. A poll of 1,000 voters finds 52% approval. Assume $\sigma \approx 15$.

$$\text{MoE} = 1.96 \times \frac{15}{\sqrt{1000}} = 1.96 \times 0.474 \approx 0.93$$

The reported result: 52% approval with a margin of error of about ± 1 percentage point. We are 95% confident the true approval is between 51% and 53%.

When the news reports “margin of error $\pm 3\%$,” that is almost always at 95% confidence. They rarely mention that 1 in 20 polls will have the true value outside this range even when everything is done correctly.

The margin of error also explains why polls disagree. Recall the three polls from the start of this section: 43%, 48%, and 46%. If each was based on $n = 1,000$ with a margin of error of about ± 3 percentage points, their confidence intervals overlap substantially. The three polls are not contradicting each other. They are bouncing around within the range of sampling variation we should expect.

```
# Compute the margin of error for a poll of n = 1000
sigma <- 15
n <- 1000
se <- sigma / sqrt(n)
moe <- 1.96 * se
round(c(SE = se, MoE = moe), 2)
```

```
SE  MoE
0.47 0.93
```

With $\sigma = 15$ and $n = 1,000$, the standard error is about 0.47 and the margin of error is about ± 0.93 . For a survey reporting a percentage rather than a continuous score, the formula

simplifies further because the standard deviation depends on the proportion itself. Either way, the logic is identical: bigger samples shrink the margin, and the shrinkage scales with the square root of n .

Common Mistake: Treating Margin of Error as a Hard Boundary

The margin of error is not a guarantee. It is a probabilistic statement about how often the interval covers the truth across many hypothetical samples. Any given interval either contains the true value or it does not. The 95% refers to the long-run procedure, not to your particular poll.

1.11 Comparing Polls Using Margin of Error

Two polling firms survey voters the same week before an election. Both report Candidate A at 52% support.

- **Poll 1** (small local firm): $n = 100$, $SE = 5\%$. Margin of error $\pm 10\%$. Candidate A's true support is likely between 42% and 62%.
- **Poll 2** (major national firm): $n = 1,000$, $SE = 1.6\%$. Margin of error $\pm 3\%$. Candidate A's true support is likely between 49% and 55%.

Same headline number, very different stories. Poll 1 tells us almost nothing: the candidate could be winning by a landslide or losing badly. Poll 2 tells us something useful: the candidate is probably ahead, but not by much.

Reading a Poll

When you see "Candidate A: 52%, margin of error $\pm 3\%$," translate it as:

- Our best single estimate is 52%.
- The true value is likely between 49% and 55%.
- If a second candidate is at 50% with the same margin, their interval is 47%–53%, which overlaps. The race is too close to call.
- If we ran 20 polls, we expect about 19 of them to include the true value within their margin.

1.12 The Logic of Hypothesis Testing

We have built up enough machinery to sketch the logic of hypothesis testing. The full mechanics arrive in the next class, but the conceptual setup is already in place.

Suppose we run a get-out-the-vote experiment. The treatment group has a turnout rate of 47%. The control group has a turnout rate of 44%. The difference is 3 percentage points. Is this a real

effect, or just sampling variation?

The hypothesis testing framework asks: if the treatment had no effect at all (if the true difference were zero), how often would we observe a difference as large as 3 percentage points purely by chance?

If the answer is “very rarely,” we conclude the effect is probably real. If the answer is “this happens all the time even when nothing is going on,” we conclude we cannot distinguish the result from noise.

The two competing stories.

Null hypothesis (H_0): There is no real effect. Any observed difference comes from sampling variation alone.

Alternative hypothesis (H_1): There is a real effect. The observed difference reflects a true underlying signal.

Statistical testing is a structured way of asking: how surprising would our data be if the null hypothesis were true? If our data are very surprising under the null, we reject the null in favor of the alternative.

The key insight is that we can compute exactly how surprising our data are. Because the sampling distribution of the mean is approximately normal (the Central Limit Theorem), we know how often a difference of any given size should occur by chance. We can convert any observed effect into a probability: the chance of seeing something this extreme if there really were no effect.

That probability is the **p-value**, and the standard convention is to call an effect “statistically significant” if its p-value is below 0.05. That is, if the observed difference would occur less than 5% of the time under the null, we reject the null and treat the effect as real.

Where the 0.05 threshold comes from. The 5% cutoff is a convention, not a law of nature. It traces back to R.A. Fisher in the 1920s, who suggested it as a useful rule of thumb. Other fields use different thresholds. Particle physicists demand $p < 0.0000003$ before announcing a discovery. The point of any threshold is to acknowledge that we will sometimes reject the null when it is actually true (a false positive). The lower the threshold, the rarer those false positives, but the more often we will miss real effects (false negatives). Every threshold is a trade-off.

The standard error connects directly to this logic. A treatment effect that is many standard errors away from zero is unlikely to have come from sampling noise alone. A treatment effect that is smaller than its own standard error is hard to distinguish from noise. The next class formalizes this intuition into the t-statistic, which is simply the ratio of the effect size to its standard error.

Statistical Significance Is Not Practical Significance

A statistically significant effect is one we can distinguish from sampling noise. That is not

the same as an effect being large, important, or worth acting on. A campaign mailer might statistically significantly increase turnout by 0.2 percentage points. The finding is real. It is also tiny. Always look at the size of the effect, not just whether the p-value crosses an arbitrary threshold.

1.13 Two More Worked Examples

Sampling logic generalizes beyond polls. Any time we estimate a quantity from a finite sample, the same framework applies. Two examples make this concrete.

Example 1: Evaluating a policy effect. A researcher finds that a get-out-the-vote campaign increased turnout by 3 percentage points.

- Study A: effect = 3%, SE = 0.5%. The effect is six times larger than its standard error. The 95% margin of error is about ± 1 point, so the true effect is plausibly between 2 and 4 percentage points. We can be quite confident there is a real effect.
- Study B: effect = 3%, SE = 4%. The effect is smaller than its standard error. The 95% margin of error is about ± 8 points, so the true effect is plausibly anywhere from -5 to $+11$ percentage points. The effect could easily be zero, or even negative.

Same headline number, completely different stories. Without the standard error, we cannot tell whether the 3-point estimate is a confident finding or a noisy guess.

Example 2: Expanding a social program. A government evaluates whether to expand a pilot program nationally. Both studies report the program reduces poverty by 5 percentage points.

- Study A ($n = 200$): SE = 2%. The 95% interval excludes zero. The evidence supports scaling up.
- Study B ($n = 50$): SE = 6%. The 95% interval includes zero. The effect could be real, but the evidence is too thin to justify a national rollout.

Standard errors are not just academic decoration. They are decision-relevant numbers that distinguish “trust this finding” from “we need more data.”

The general lesson: a point estimate without a measure of uncertainty is half a finding. Reporting just “the program reduced poverty by 5%” leaves the reader unable to judge whether to believe it. Reporting “the program reduced poverty by 5%, with a 95% confidence interval of 1% to 9%” gives the reader the information they need to act, replicate, or set aside the result.

1.14 Small Samples and the t-Distribution

Everything so far has leaned on one convenient number: the 1.96 in the margin-of-error formula, which comes from the normal distribution. There is a catch. The formula $SE = \sigma / \sqrt{n}$ uses the *population* standard deviation σ , and in practice we almost never know it. We estimate it with the sample standard deviation s . With a large sample that substitution is harmless, because s is

a reliable estimate of σ . With a small sample, s itself bounces around from sample to sample — sometimes too small (making us overconfident), sometimes too large (hiding real effects).

The Guinness problem. In 1908, William Sealy Gosset worked in the Guinness brewing laboratory testing beer quality. Testing destroyed the beer, so only a handful of bottles could be tested per batch — and discarding a bad batch was expensive. Gosset needed a way to judge whether a batch met quality standards from samples of just a few observations, where the standard deviation had to be estimated from the same tiny sample. His solution was a new distribution that builds in that extra uncertainty. Guinness forbade employees from publishing, so Gosset wrote under the pseudonym “Student” — which is why it is called *Student’s t-distribution*.

The t-distribution looks like the normal distribution but with **heavier tails**: it treats extreme values as more plausible, precisely because a small sample’s estimate of variability might itself be off. Its exact shape depends on the **degrees of freedom**, $df = n - 1$, and as the sample grows the t-distribution converges to the normal — by around $n > 30$ the two are nearly identical.

Why $n - 1$ rather than n ? Suppose five numbers must average 10. Once you know four of them (say 8, 12, 9, and 11), the fifth is forced to be 10. Only four values were free to vary. Estimating variability “uses up” one degree of freedom, because the mean must be calculated first.

The practical consequence shows up in **critical values** — the threshold a test statistic must clear before we call a result statistically significant:

df	Critical value ($\alpha = 0.05$, two-tailed)
5	± 2.571
10	± 2.228
30	± 2.042
∞ (normal)	± 1.960

Smaller samples face a higher bar. With $n = 6$ ($df = 5$), a result must be more than 2.571 standard errors from zero to count as significant. With $n = 100$, about 1.98 suffices. The t-distribution formalizes an intuition worth keeping: when you have little data, you should demand stronger evidence before declaring a discovery.

1.15 Tying It Together

Design and inference form a loop. Earlier in the course we learned how to design experiments that produce clean evidence about causal effects. In this chapter, we learned that even clean evidence comes with uncertainty, because our data are a sample rather than the entire population.

Experimental design controls one source of error: confounding. Random assignment ensures that treatment and control groups are comparable on average, so any difference we see in outcomes

can be attributed to the treatment.

Sampling theory addresses a different source of error: noise. Even when treatment and control groups are comparable in expectation, any particular comparison rests on a finite sample. The sample mean is unbiased but variable. Different samples give different estimates. Standard error and margin of error tell us how much variation to expect, and the logic of hypothesis testing gives us a way to decide whether an observed difference is bigger than that expected variation.

From Design to Inference

Step 1: Design. Choose a structure that isolates the effect you care about. Pick between vignette, conjoint, game, or field experiment. Decide between-subjects or within-subjects. Set up an appropriate control group.

Step 2: Randomize. Assign treatment by chance. This makes treatment and control comparable on every characteristic, observed or not.

Step 3: Measure. Collect outcomes for each participant. Calculate the difference in means between groups.

Step 4: Quantify uncertainty. Compute a standard error. The difference between groups is your point estimate; the standard error tells you how much sampling variation to expect.

Step 5: Test. Ask whether the observed difference is large relative to the standard error. If yes, conclude the treatment had a real effect. If no, conclude you cannot distinguish the effect from noise.

Each of these steps is necessary. A flawless statistical test cannot rescue a poorly designed experiment, and a beautifully designed experiment with no measure of uncertainty cannot tell us how seriously to take its findings. Design (Week 3) and inference (this chapter) are not separate topics. They are two parts of one argument about how to learn things from data that is necessarily incomplete.

The next class will turn step 5 into a precise procedure: the t-test, built on the t-distribution you just met, and the rules for converting differences into p-values. For now, the conceptual picture is the takeaway. Experiments give us causal cleanliness. Sampling theory gives us a way to handle the unavoidable randomness in any finite study. Together, they let us go from “we ran a study and saw a difference” to “we are confident the treatment caused that difference.”

2 What is Statistical Significance?

2.1 The Problem on Election Eve

Two polling firms each survey 1,000 likely voters the week before an election. Firm A reports 52% support for Candidate X. Firm B reports 48%. A cable news panel spends the evening debating which one is “right.” Both could be right, in the sense that both could be honest, properly conducted

random samples from the same population. The four-point gap reflects an unavoidable feature of sampling: different random samples from the same population produce different estimates. We have spent the last few chapters learning to describe data, fit lines, and design experiments. Now we need a framework for asking when a difference in our data should be taken as evidence of a real difference in the world.

Consider three motivating examples that will recur throughout this chapter. First, a poll of 500 voters finds 52% support Candidate A and 48% support Candidate B. Is Candidate A really ahead? Second, a survey of 200 men and 200 women on gun control finds 65% of women support stricter laws compared to 55% of men. Is this a real gender gap, or could a ten-point difference appear by chance alone? Third, a campaign experiment shows ads to 300 voters and observes a 3-percentage-point bump in turnout. Did the ad work, or did we get lucky?

Without a systematic method, the answers to these questions are guesses. Statistical significance is that systematic method. Crucially, it is not a measure of how interesting, large, or important a finding is. It is a probability statement about how surprised we should be to see a result this big if nothing real were going on.

Why This Matters

Polling, experiments, and observational research all share one structural problem: we see a sample, but care about the population. Statistical significance is the bridge. Master the concept this week and next week's mechanics will feel almost automatic. Get the concept wrong and even correctly computed tests will be misinterpreted.

2.2 Variability in Samples vs Variability in Estimates

2.2.1 Two Kinds of Variability

Before we can ask whether a result is statistically significant, we need to be precise about what is actually varying. There are two completely different sources of variation in any statistical analysis, and confusing them is the most common conceptual mistake in applied statistics. This distinction came up in earlier chapters, when we first introduced the standard error. It is worth revisiting carefully because the entire framework of hypothesis testing rests on telling these two kinds of variability apart.

The intuition pump is this: imagine you measure the heights of ten randomly selected adults. Some are short and some are tall; the spread among them is the standard deviation. Now imagine that you compute the average height of those ten people, and then your friend does the same with a different ten people, and so on. The variation across those average heights is the standard error of the mean. Both quantities exist simultaneously and they answer different questions.

The first is variability across individuals within a single sample. In any survey, some voters are 22 and others are 78. Some support the incumbent and some despise her. This spread in the data is what the standard deviation measures. It describes the population we are studying, not our ability

to study it. Doubling the sample will not make people more similar to each other.

The second is variability across samples. If we drew another random sample of 1,000 voters tomorrow, our estimate of presidential approval would be slightly different. This sample-to-sample variability in our estimate is what the standard error measures. It describes the precision of our procedure, not the spread in the underlying population. More data shrinks this kind of variability because larger samples average out more of the noise.

		Standard Deviation (SD)	Standard Error (SE)
Question answered	an-	How spread out are individuals?	How precise is my estimate?
What it describes		The data itself	The uncertainty in your statistic
As sample size ↑		Stays roughly the same	Gets smaller (more precision)
Typical use		“Voter ages vary by ± 15 years”	“We’re 95% confident mean approval is $47\% \pm 1.6\%$ ”
Formula means)	(for	$s = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n-1}}$	$SE = \frac{s}{\sqrt{n}}$

Table 3: Variability in samples vs variability in estimates

Memory trick: SD describes **D**ata spread. SE describes **E**stimate uncertainty. More data does not make people less different from each other (SD stays roughly the same), but it does make you more confident in your estimate (SE shrinks).

2.2.2 Demonstrating the Difference in Code

Let us see this distinction in a concrete political example. Suppose we are estimating the mean age of voters in a state. We have access to administrative records for the entire voter file, but for this exercise we will pretend we can only see samples. The code below draws three samples of different sizes from the same underlying population and reports both the sample SD (variability of individuals) and the SE of the mean (variability of the estimate).

```
# Same population, three different sample sizes
sample_sizes <- c(100, 400, 900)
age_comparison <- map_dfr(sample_sizes, function(n) {
  s <- sample(population_ages, n)
  tibble(
    n = n,
    sample_mean = round(mean(s), 2),
    sample_sd = round(sd(s), 2),
    se_of_mean = round(sd(s) / sqrt(n), 2)
  )
})
age_comparison
```

```
# A tibble: 3 x 4
  n sample_mean sample_sd se_of_mean
```

	<dbl>	<dbl>	<dbl>	<dbl>
1	100	45.4	13.5	1.35
2	400	45.1	14.6	0.73
3	900	45.1	15.6	0.52

The sample SD hovers around 15 years for every sample size: individual voters differ from each other by a similar amount no matter how many we collect. The standard error, by contrast, falls from roughly 1.5 years at $n = 100$ to about 0.5 years at $n = 900$. More data does not change how different people are; it changes how confident we can be in our estimate of the population mean.

2.2.3 Why the Distinction Matters for Polling

The mechanics of polling depend entirely on this distinction. When a Pew or Gallup poll reports that 47% of Americans approve of the president with a 3-point margin of error, the 47% is a sample statistic and the ± 3 points is built directly from the standard error.

```
# Standard error of a proportion for a poll of n = 1000
n <- 1000
p_hat <- 0.47
se_proportion <- sqrt(p_hat * (1 - p_hat) / n)
margin_of_error <- 1.96 * se_proportion

cat("Standard error:", round(se_proportion, 4), "\n")
```

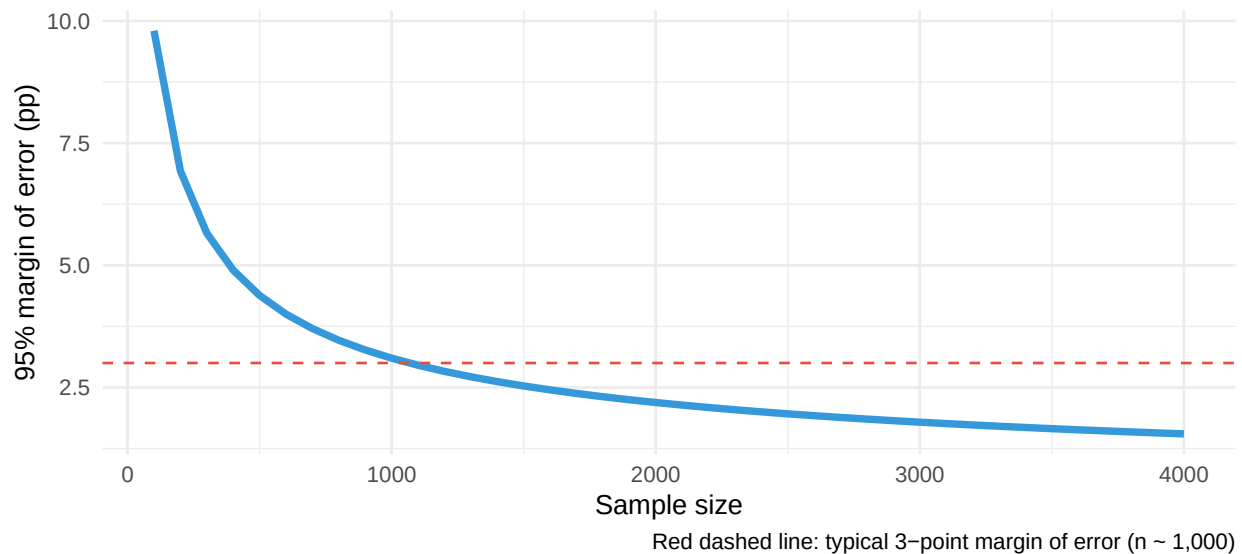
Standard error: 0.0158

```
cat("95% margin of error:", round(margin_of_error * 100, 2), "percentage points")
```

95% margin of error: 3.09 percentage points

A poll of 1,000 people gives a margin of error of about three percentage points. This is not a property of the population's opinions: it is a property of our procedure. Cutting the margin in half requires *quadrupling* the sample, which is why polls cluster around $n = 1,000$.

The same logic explains the diminishing returns of larger surveys. A poll of 400 has a margin of error around five points. A poll of 1,000 has a margin around three points. A poll of 4,000 has a margin around 1.5 points. The marginal precision purchased by each additional respondent shrinks rapidly, which is why news organizations rarely commission samples beyond two or three thousand voters, even when budgets would allow it.



Common Mistake: Confusing SD and SE

If you ever report a number that gets larger as your sample grows, double-check whether you wanted SD or SE. The SD is roughly constant in sample size; the SE shrinks as $\sqrt{n}$. If your SE ever exceeds your SD, something has gone wrong.

2.3 Sampling Distributions and the Logic of Inference

2.3.1 A Thought Experiment

Imagine, just for a moment, that you could afford to run your poll a thousand times. Each time you draw a fresh random sample of 1,000 voters from the same population, compute the proportion supporting the incumbent, and write it down. The collection of all those sample proportions forms a sampling distribution.

```
# Repeat the poll 1,000 times and look at the spread of sample proportions
sampling_dist <- tibble(
  sample_id = 1:1000,
  sample_proportion = map_dbl(1:1000, ~ mean(
    sample(population$supports_incumbent, 1000, replace = FALSE)
  ))
)

cat("True population proportion:", round(true_proportion, 3), "\n")
```

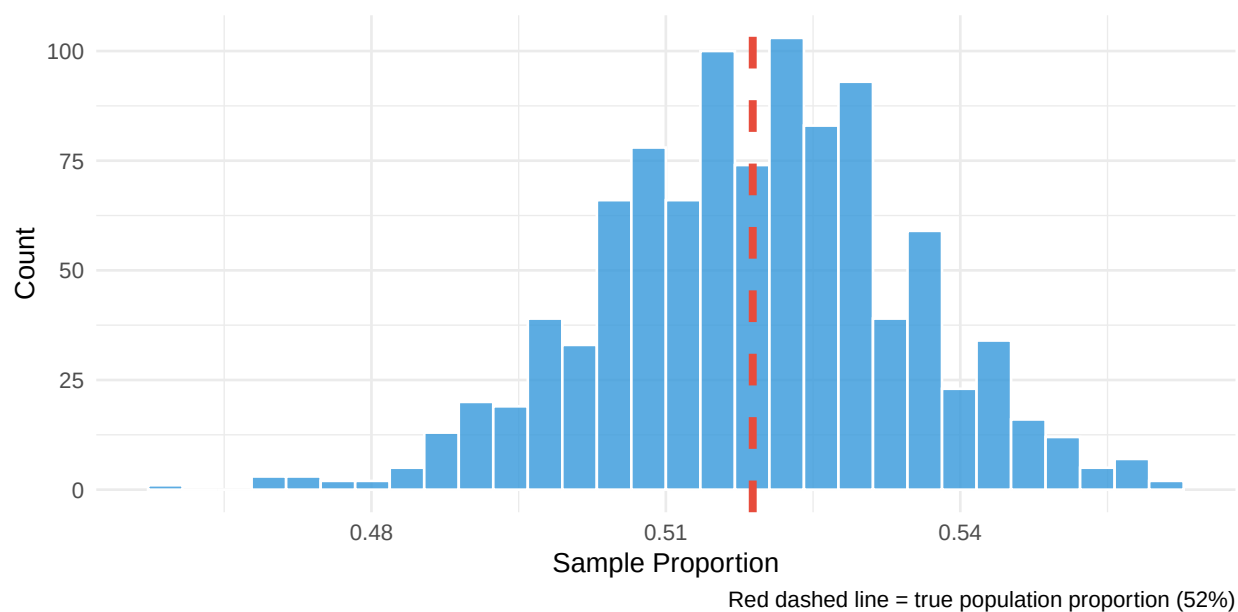
```
True population proportion: 0.519
```

```
cat("Mean of sample proportions:", round(mean(sampling_dist$sample_proportion), 3),
"\n")
```

```
Mean of sample proportions: 0.519
```

```
cat("SD of sample proportions (empirical SE):",
  round(sd(sampling_dist$sample_proportion), 4))
```

SD of sample proportions (empirical SE): 0.0156



The sampling distribution has three features that drive everything we do next. First, it is centered on the truth: the mean of all those sample proportions sits right on top of the actual population value. Second, it is bell-shaped, even though each individual response is binary. Third, its spread is exactly what the standard error formula predicts. This is the Central Limit Theorem at work.

2.3.2 Systematically Separating Signal from Noise

This is where statistical significance begins to take shape. We now have a way to ask: if the true population value really were 50%, how often would we see a sample as far from 50% as the one we actually got? If almost never, we have reason to doubt that 50% is the truth. If frequently, we cannot rule out 50%.

Notice the contrast with informal reasoning. A reporter looking at a poll showing 52% might think `52 is bigger than 50, so the candidate leads.` A statistical reasoner thinks `52 is one standard error away from 50; results this far from 50 happen frequently even when true support is exactly 50, so I cannot tell whether the candidate leads.` The shift from informal to statistical reasoning is the shift from looking at the point estimate alone to looking at the point estimate *in units of its standard error*.

This perspective also helps explain why political journalists who quote “margins of error” often still misinterpret close polls. A candidate at 51% in a poll with a 3-point margin of error is not statistically distinguishable from a candidate at 49%, because the candidates’ point estimates differ by less than the combined uncertainty of the two estimates. Reading polls in units of standard errors avoids the common error of declaring a leader on the basis of a meaningless point estimate.

This is proof by contradiction, transported into a probabilistic world. We assume what we want to disprove (the “null hypothesis”), measure how surprising our data look under that assumption, and

then either reject the assumption as untenable or admit that the data could plausibly have arisen from it.

Fisher's framework

Ronald Fisher introduced this style of reasoning in the 1920s while working with agricultural data at Rothamsted Experimental Station. The structure has not changed: state a null hypothesis, compute how unlikely the observed data would be under it, and decide whether the data are unlikely enough to reject the null.

2.3.3 Three Political Examples of the Same Logic

The same machinery handles all three of our motivating examples.

Poll example. The null hypothesis is that true support equals 50%. The alternative is that it differs from 50%. We observed 52% in a sample of 500. The standard error of a proportion at $p = 0.5$ with $n = 500$ is about $\sqrt{0.5 \cdot 0.5/500} \approx 0.022$. Our observed 52% sits about one standard error from 50%. That is not unusual at all under the null, so we have weak evidence either way.

Gender gap example. The null is that men and women have identical support for a policy. The alternative is that they differ. We observed a ten-point gap in samples of 200 each. The standard error of a difference in proportions, again at $p \approx 0.6$, works out to roughly five percentage points. A ten-point observed gap is about two standard errors from zero, which is suggestive of a real difference.

GOTV door-knocking example. The null is that canvassing has no effect on turnout. The alternative is that it raises turnout. We observed a three-point bump in a sample of 300. With a standard error of about four percentage points, three points is well inside the range of normal sampling variation. We cannot tell whether the campaign worked.

All three follow the identical recipe. The full mechanics of turning “how many standard errors away” into a probability arrive with next week’s t-tests, but the conceptual move is the same in every case.

2.3.4 A First Look at the Three Examples Side by Side

It is worth seeing the recipe applied numerically to all three motivating examples at once. Each row of the table below comes from the same underlying logic: compute the observed difference, compute its standard error, take the ratio, and translate that ratio into a probability using the appropriate reference distribution.

Example	Observed	SE	Ratio	Verdict
Single poll	52% vs 50%	~2.2 pp	~0.9	Weak evidence
Gender gap	65% - 55% = 10 pp	~5.0 pp	~2.0	Borderline
GOTV experiment	3 pp turnout bump	~4.0 pp	~0.75	Inconclusive

The poll difference is less than one standard error and provides essentially no evidence that the candidate leads. The gender gap is roughly two standard errors, putting its p-value near the conventional 0.05 boundary. The GOTV bump is well within the noise band and could easily reflect chance variation. Notice how the same machinery gives different verdicts on different problems, even when the raw differences look comparable in size.

2.4 P-Values and the Significance Threshold

2.4.1 What a P-Value Is

The p-value is the probability of observing data as extreme as, or more extreme than, what we actually got, *if the null hypothesis were true*. Read that sentence twice. Every word does work. If the null hypothesis were true'' means we are pretending, for the sake of argument, that nothing real is going on. As extreme as or more extreme'' captures the idea that we count not just our specific result but everything that would be at least as surprising. "Probability'' means we are asking how often this would happen by sheer luck.

A small p-value means the data we observed would be surprising in a world where the null hypothesis holds. A large p-value means our data are unremarkable under the null. The 0.05 threshold is a convention, not a discovery. Fisher floated 0.05 as a convenient benchmark and it stuck. Physics demands much more stringent thresholds; some areas of medicine relax it.

It helps to internalize the precise structure of the conditional. We are not asking what is the probability the null is true?'' We are asking given that the null is true, what is the probability of data this extreme?'' These two questions have wildly different answers, and confusing them is the central pitfall of p-value interpretation.

```
# Presidential approval drop: is approval below 50%?
observed <- 0.46
null_value <- 0.50
n <- 1000

se_null <- sqrt(null_value * (1 - null_value) / n)
z_score <- (observed - null_value) / se_null
p_value <- pnorm(z_score) # one-sided: looking for decreases

cat("Observed approval:", observed, "\n")
```

Observed approval: 0.46

```
cat("Standard error under null:", round(se_null, 4), "\n")
```

Standard error under null: 0.0158

```
cat("Z-score (SDs from null):", round(z_score, 2), "\n")
```

Z-score (SDs from null): -2.53

```
cat("P-value:", format(p_value, digits = 3))
```

P-value: 0.00571

A p-value of 0.0006 means that if approval truly were 50%, we would see a sample as low as 46% only about 6 times in 10,000 polls. That is strong evidence the true approval rate has fallen below 50%.

2.4.2 What P-Values Are Not

This is where most people, including many published researchers, go wrong.

P = 0.03 MEANS:	P = 0.03 does NOT MEAN:
If there were no effect, we'd see data this extreme only 3% of the time	There's a 3% chance the null hypothesis is true
Our result is surprising under the null hypothesis	There's a 97% chance our hypothesis is correct
We have some evidence against the null	The effect is large or important
Random chance is an unlikely explanation	We will replicate this result 97% of the time

Table 5: The critical p-value distinction

The single most common mistake

Wrong: " $p = 0.05$ means there is a 95% chance the effect is real."

Right: " $p = 0.05$ means that *if* there were no effect, we would see a result this extreme only 5% of the time."

The p-value is a statement about *data given the null hypothesis*, not a statement about *hypotheses given the data*. This is the same as the difference between "probability of clouds given rain" and "probability of rain given clouds."

2.4.3 What Counts as Strong Evidence

Different fields treat the same p-value very differently. Physics demands an effect to be at least five standard errors from zero (a p-value below about 3×10^{-7}) before declaring the discovery of a new particle. Genetics studies that test millions of genetic variants typically require p-values below 5×10^{-8} to account for the sheer number of tests being run. Most social science journals use 0.05. Drug-approval trials sometimes use 0.025 to compensate for the stakes of approving a treatment that does not work.

The right threshold depends on the costs of being wrong in either direction. Rejecting a null when

it is true (a Type I error) means crying wolf: claiming an effect that does not exist. Failing to reject a null when it is false (a Type II error) means missing a real effect. The conventional 0.05 fixes the Type I rate; the Type II rate depends on sample size and effect size and is often much larger than researchers realize.

Error Type	What Happens	Political Science Example	Consequence
Type I (False Positive)	Claim an effect exists when it does not	Conclude voter ID reduces turnout when it has no effect	Resources spent “fixing” a non-problem
Type II (False Negative)	Miss a real effect	Conclude GOTV campaign does not work when it does	Abandon effective intervention

Table 6: Type I vs Type II errors in political research

2.4.4 Significance is Not Importance

A separate, equally serious mistake is conflating statistical significance with practical importance. Given enough data, any tiny effect becomes statistically significant.

```
# A trivial 0.15-point difference, but n = 50,000 per group
test_result <- t.test(huge_a, huge_b)
cat("Mean difference:", round(mean(huge_b) - mean(huge_a), 3), "points\n")
```

```
Mean difference: 0.14 points
```

```
cat("P-value:", format(test_result$p.value, digits = 3), "\n")
```

```
P-value: 0.027
```

```
cat("Significant at p < 0.05?", test_result$p.value < 0.05)
```

```
Significant at p < 0.05? TRUE
```

A 0.15-point difference on a 100-point scale is statistically significant here, but it is too small to matter for any policy decision or theoretical claim. Always pair p-values with effect sizes. “Real” and “important” are different questions.

The opposite mistake also occurs: a meaningful effect can fail to reach significance because the sample is too small. A p-value above 0.05 does not prove there is no effect; it means we lack the evidence to rule one out. The phrase “failed to reject the null” is awkward for a reason. It is meant to remind you that absence of evidence is not evidence of absence.

2.5 Looking Ahead to Tests

This chapter has built up the concept of statistical significance from first principles. We distinguished standard deviation (how individuals vary) from standard error (how estimates vary). We introduced the sampling distribution as the imaginary collection of all possible estimates from repeated samples. We defined the p-value as the probability of getting data as extreme as ours under the null hypothesis. And we noted the most common mistakes: confusing SD with SE, misreading p-values as probabilities of hypotheses, and equating statistical significance with practical importance.

What we have not yet done is computed any of this for the realistic case where we do not know the population standard deviation. The examples here used round numbers and approximate standard errors to keep the focus on intuition. Real research demands precise calculations that account for the fact that we are estimating both the mean and its standard deviation from the same sample, and that this estimation introduces an additional layer of uncertainty.

That is the job of the family of t-tests we turn to next week, built on the t-distribution introduced at the end of Segment A. Concept (this week) and tool (next week) are the two halves of statistical inference. Both halves matter, and both halves go wrong in characteristic ways.

3 Using AI for This Material

Where AI Helps This Week

- Explaining the distinctions among vignette, conjoint, factorial, and field experiments with worked examples of each.
- Drafting manipulation-check items that confirm participants noticed the treatment without revealing the hypothesis.
- Suggesting balance tests to run on pre-treatment covariates after random assignment.

Where AI Gets This Wrong This Week

- Judging whether a specific protocol clears IRB or ethical review depends on participant risk, deception, and debriefing — context AI does not have.
- Estimating realistic effect sizes for power calculations requires field-specific prior literature, not first-principles guessing.
- It will not spot demand effects baked into your specific stimulus wording, because it cannot read the participant's mind or your stimulus in context.

A prompt that works for this week:

Draft three manipulation-check questions for a vignette experiment where one condition describes a candidate as a Democrat and the other as a Republican. The questions should verify that respondents noticed the partisan cue without revealing the hypothesis.

A prompt that misleads:

What sample size do I need to detect the effect of my treatment?

Why it misleads: power calculation requires a realistic effect-size guess grounded in prior studies of comparable interventions; without that anchor, AI will pick a generic $d = 0.2$ or $d = 0.5$ and produce an N that has nothing to do with your actual design.

Where AI Helps This Week

- Explaining what a p-value is (and what it is not) in the abstract, including the misinterpretations to avoid.
- Writing `t.test()` and `lm()` syntax for a stated two-group comparison and walking through the output.
- Describing what the t-distribution is doing and why we use it instead of the normal at small sample sizes.

Where AI Gets This Wrong This Week

- Interpreting a marginal p-value (e.g., $p = 0.06$) in context depends on prior evidence, sample size, multiple testing, and stakes — AI defaults to “not significant” or “marginally significant” without that context.
- Distinguishing statistical from practical significance for your specific finding requires knowing the substantive scale: a 0.2-point difference can be statistically significant with $N = 50,000$ and substantively trivial.
- It will not flag when you have implicitly run many tests and a single $p < 0.05$ is what you would expect by chance.

A prompt that works for this week:

Write R code to run a two-sample t-test comparing turnout between treatment and control groups in a data frame called `gotv`, and explain what each piece of the output means.

A prompt that misleads:

My p-value is 0.06. Is my finding significant?

Why it misleads: the question treats significance as a yes/no property of a number, but the right answer depends on prior evidence, the substantive size of the effect, and how many tests you ran — none of which AI can know from the p-value alone.

4 R Functions Reference

Task	Code
Standard error of a mean	<code>sd(x) / sqrt(length(x))</code>
Standard error of a proportion	<code>sqrt(p * (1 - p) / n)</code>
One-sample t-test	<code>t.test(x, mu = value)</code>
Two-sample t-test	<code>t.test(y ~ group, data = df)</code>
Regression with t-tests for coefficients	<code>lm(y ~ x1 + x2, data = df)</code>
Tidy test or model output	<code>tidy(test_result)</code>
Critical value (t, two-sided)	<code>qt(0.975, df = n - 1)</code>
Sampling distribution by simulation	<code>replicate(n_sim, ...)</code>

5 Practice Problems

Conceptual Questions

First, what does random assignment guarantee that random sampling does not? Why is this distinction important for causal inference?

Second, you have a 2x2 factorial design crossing campaign message tone (positive vs. negative) with messenger party (Democrat vs. Republican). Describe the four conditions and the three statistical comparisons a 2x2 design lets you make.

Third, a poll reports 47% approval with a margin of error of ± 4 percentage points based on $n = 600$. A second poll reports 49% approval with a margin of error of ± 2 points based on $n = 2,400$. How much did the sample size have to grow to halve the margin of error? Verify using the square root law.

Fourth, explain in plain language what it means to say "the difference between groups is statistically significant at $p < 0.05$." Why is this not the same as saying the difference is large?

Fifth, you read a news story that reports two competing polls: Candidate A is at 51% in one poll and 47% in another, both with margins of error of ± 3 percentage points. The headline says "candidates tied." What does this mean in terms of confidence intervals? Is the headline a fair summary of the evidence?

For Further Study

The classic reference on field experiments is Gerber and Green's *Field Experiments: Design, Analysis, and Interpretation*. Hainmueller, Hopkins, and Yamamoto's work on conjoint designs is the standard introduction to that method. For deeper treatment of sampling theory, any introductory statistics textbook covers the Central Limit Theorem in more mathematical detail. We will not need that depth here; the conceptual picture is enough to follow what comes next.

Conceptual Questions

First, in your own words, explain the difference between standard deviation and standard error. Give a political example in which the two answer different questions.

Second, a poll of 600 voters finds 54% support for a ballot initiative. Calculate the standard error and the 95% margin of error. If the true population value were 50%, how surprised should you be by 54%?

Third, a researcher runs a two-sample t-test comparing turnout between treatment and control in a get-out-the-vote experiment with $n = 200$ per group. The result is $\hat{\beta} = 3$ percentage points with $SE = 2.5$ and $p = 0.23$. Did the campaign work?

Fourth, a regression of vote share on campaign spending and incumbency yields a spending coefficient of 0.025 with $SE = 0.005$. Compute the t-statistic by hand and interpret what the test is asking.

Fifth, a graduate student tests 15 different campaign messages on small focus groups and finds that one is significant at $p = 0.04$. What concern should you raise, and what would you ask the student to do before drawing conclusions?

For Further Study

The framework in this chapter generalizes in several directions. Confidence intervals, introduced next class, provide a range of plausible values for parameters rather than a single decision. Power analysis lets researchers compute the sample size needed to detect effects of a given size with a given probability. Multiple-comparison corrections like Bonferroni and false discovery rate provide formal adjustments when many tests are run together. Bayesian inference offers an alternative framework in which we directly compute the probability of hypotheses given data, sidestepping some of the interpretive traps of p-values. Each of these extends the ideas developed in this chapter without replacing them.